RESEARCH PROTOTYPE

GlycoTwin

Two-hour glucose forecasting with baseline clinical and wearable context

A retrospective digital-twin-oriented proof of concept using CGMacros

NOT CLINICALLY VALIDATED • NOT FOR DIAGNOSIS OR CARE

45 participants | 45,447 forecast windows | grouped participant evaluation

Why a two-hour forecast?

A high-burden problem

ICMR-INDIAB estimated 101 million people in India had diabetes in 2021 (weighted prevalence 11.4%). The project explores earlier visibility into possible high glucose.

A bounded research question

At each 15-minute replay point, estimate whether Libre CGM glucose will be ≥180 mg/dL at t+120 minutes. The horizon is challenge-aligned and exploratory—not a clinical standard.

Who would review it?

Concept user: diabetologist/endocrinologist during follow-up for a person with established T2D. Review the trace and context, verify with the patient, then follow usual care processes.

No medication recommendation, diagnosis, triage action, or patient alert is validated.

The “alert” is a research interface concept only.

180 mg/dL is a common CGM upper-range boundary; this project does not validate a 2-hour intervention window.

The method may transfer; these results do not

SOURCE COHORT: UNITED STATES

45	~10d avg; <=19d max	14	16	15
participants	multimodal records	diabetes-range HbA1c	prediabetes-range HbA1c	healthy-range HbA1c

What can transfer

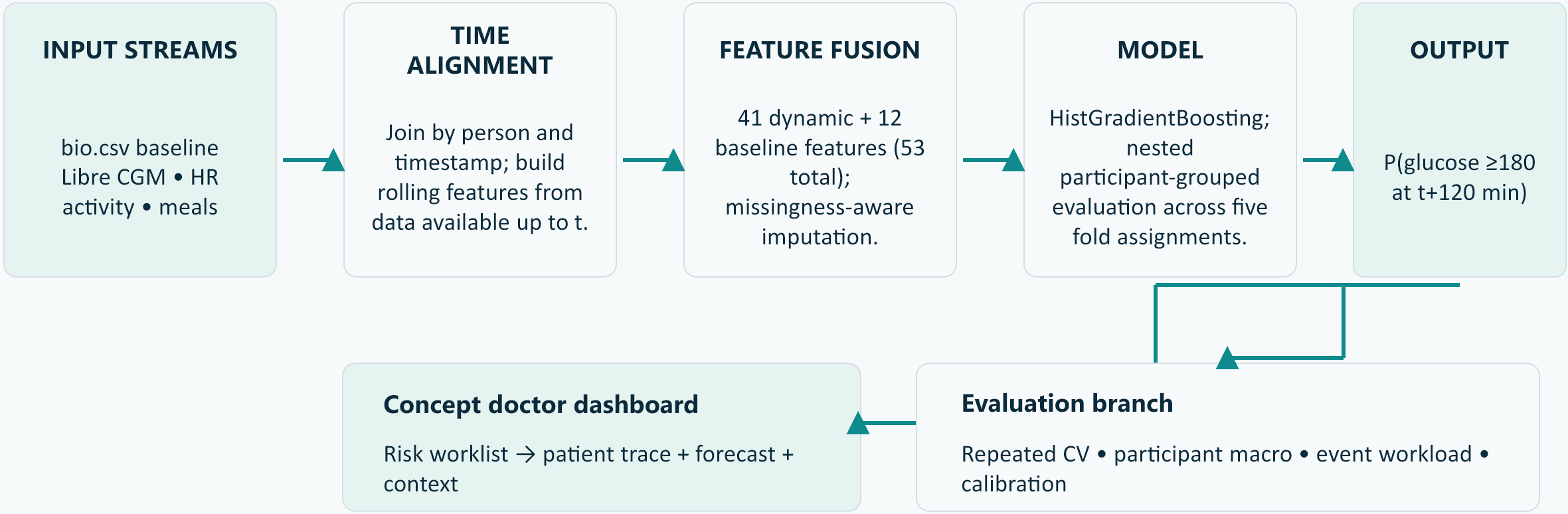
The workflow: align streams by participant and time, fuse available context, forecast a future CGM threshold, and evaluate on held-out people.

What cannot transfer yet

Learned risk levels, calibration, subgroup results, and performance. India-specific use needs representative Indian data, local retraining, prospective validation, and threshold evaluation.

Cohort strata in this analysis use baseline HbA1c bands, not independent chart diagnoses.

Architecture: from multimodal data to a forecast



Sleep and HRV are possible future inputs only; neither is included in this CGMacros model.

Model and evaluation design

Prediction target

At 15-minute replay points, forecast glucose ≥180 mg/dL at +30, +60, and +120 min.

53 fused features: 41 dynamic + 12 baseline EHR.

Imputation and model fitting stay within training participants.

Selected model

HistGradientBoosting classifier / regressor

Four representative settings; max_iter 300–1,000 (iterations, not epochs).

Inner grouped AP / MAE selection; outer folds hold out people. Five distinct assignments.

45,447 OOF WINDOWS

2,845 POSITIVE EVENTS

43/45 EARLY-DATA ELIGIBLE

Repeated grouped evaluation: assignment spread

Pooled PR-AUC

0.370

Seed range 0.358–0.382

Pooled ROC-AUC

0.878

Seed range 0.855–0.888

Macro ROC-AUC

0.741

Seed range 0.728–0.751

Participant-level ranking

Event prevalence is 6.26% (constant-prevalence AP 0.063). Mean pooled PR-AUC is 0.370; participant-macro AP is 0.243. Pooled ROC-AUC exceeds macro ROC-AUC because pooled ranking also rewards ranking between people.

Horizon and alert trade-off

30-min ranking is strongest; 120-min top-5% detects 122/384 episodes in 13/45 people, with 1.14 false alerts per approximate person-day and 90-min median lead. Merge/cooldown rules are exploratory; no horizon is clinically selected.

Five distinct participant assignments; shown ranges are split-to-split spread, not confidence intervals. The cohort is small and US-based; no external validation.

Current seed-42 HbA1c strata

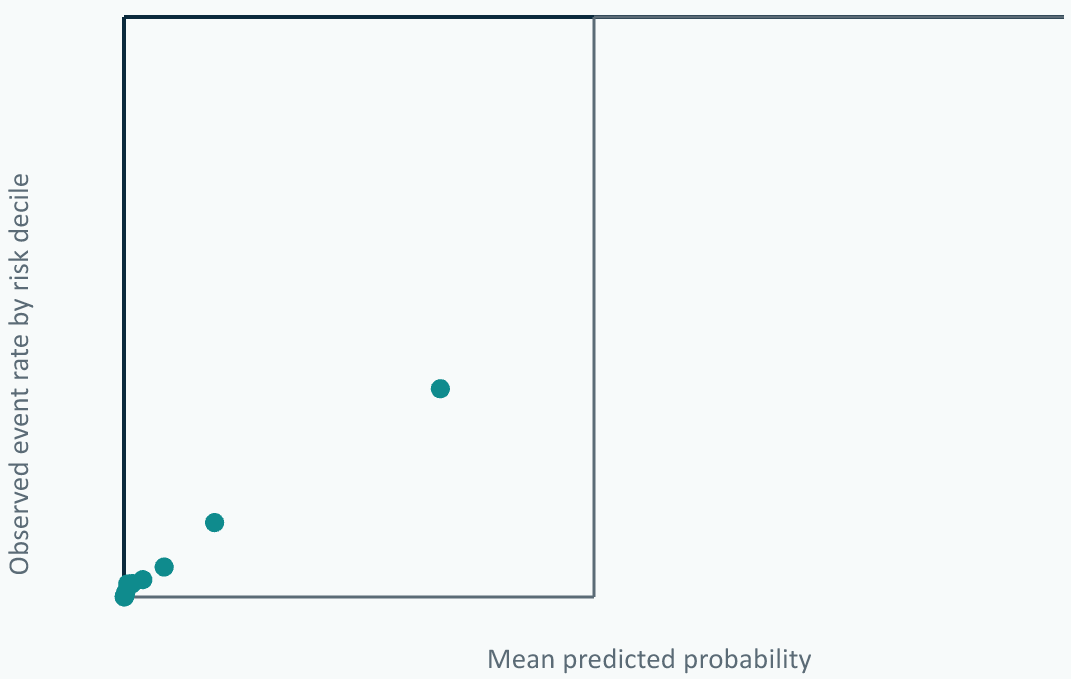
Analytic HbA1c band	People	Events (n+)	Rate	PR-AUC	ROC-AUC
Healthy-range <5.7%	15	106 (2)	0.7%	0.044	0.893
Prediabetes-range 5.7–6.4%	16	439 (13)	2.8%	0.164	0.832
Diabetes-range ≥6.5%	14	2,300 (14)	15.9%	0.430	0.803

Interpret with caution

n+ = participants with at least one event. AP baselines are 0.7%, 2.8%, and 15.9%; healthy-range events come from just two people.

Cluster-bootstrap 95% CIs (PR/ROC): healthy 0.000–0.264/0.343–0.944; prediabetes 0.037–0.274/0.739–0.891; diabetes 0.266–0.586/0.738–0.851.

Calibration and participant-level uncertainty



Calibration is descriptive

Ten quantile bins use seed-42 held-out predictions. This single-assignment curve is descriptive; low event counts and the small US cohort make calibration uncertain.

Uncertainty unit = participant

Participant-cluster bootstrap 95% CI (seed 42): pooled PR-AUC 0.216–0.529; ROC-AUC 0.825–0.931. Intervals condition on fixed OOF scores and omit model-refit uncertainty.

No post-hoc calibration; no external cohort.

Concept screen 1 / Doctor worklist

GLYCOTWIN / FOLLOW-UP WORKLIST NOT CLINICAL

Research replay · sample patient identifiers · scores below are illustrative

PATIENT	GROUP	LATEST CGM	2H SCORE*	REVIEW
P-023	T2D-range	168 mg/dL	**62%**	Open
P-011	Prediabetes	142 mg/dL	**41%**	Open
P-006	T2D-range	131 mg/dL	**29%**	Open

1 Sort by model score for review in this prototype; this is not clinical urgency or triage.

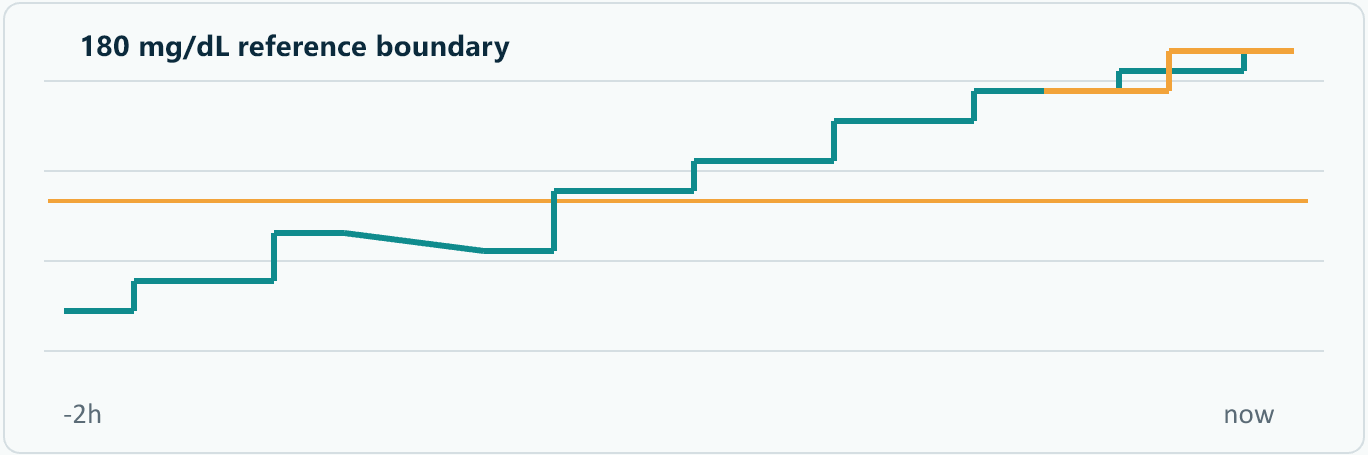
2 Select a row to inspect the glucose trace, forecast window, and available context.

***Illustrative UI values—not actual patient predictions. No real patient data shown.**

Concept screen 2 / Single-patient review

PATIENT P-023 / CONCEPT VIEW NOT CLINICAL

Recent CGM trace with illustrative t+120 forecast



Illustrative 2-hour score

62%

Context available

Baseline profile • recent meal • HR/activity summaries

Review sequence

1 Check data quality and sensor context

2 Ask about meals, activity, and symptoms

3 Apply usual clinical assessment and care

Forecast and patient values are illustrative. This is a conceptual UI, not a clinical display.

What must happen before real-world use

Current limitations

- 45-person US retrospective cohort
- No Indian or external validation
- No prospective clinical or utility study
- No treatment-response model
- No sleep/HRV inputs
- Alert episode merge/cooldown rules are exploratory
- Calibration is descriptive only
- Personalization: fixed 72h; 43/45 early-data eligible

Responsible validation path

1 Fixed 72h timeline: 43/45 qualify by early data only. Pooled/macro AP: population .374/.240; early calibration .538/.240 (macro unchanged; 20 fit); early-data adaptation .523/.251; person-only .340/.140.
2 Regression MAE on all 45: population 20.82, early-data adaptation 18.10, person-only 20.59 mg/dL.
3 Collect representative Indian cohort and treatment-context data; validate prospectively.
4 Clinicians define event merge, cooldown, warning/action window, and acceptable workload.
5 Review safety, equity, drift, ethics, privacy, security, and regulation.

Privacy review: assess India DPDP obligations; HIPAA applies only when covered entities/business associates and data are in scope.

Method and fair comparison

The evaluation separates participant generalization, personalization, and exploratory workload decisions.

Repeated grouped CV

Five distinct participant assignments; outer participants remain held out.

Inner participant-grouped search; cutoffs selected from training-participant OOF scores.

Report pooled and participant-macro metrics because they answer different questions.

Personalization

Fixed first 72 hours for adaptation/training; 43/45 participants qualify (≥200 early labeled intervals).

Population + early data in model fit: pooled PR-AUC 0.523; macro PR-AUC 0.251.

Regression pooled MAE: 20.82 → 18.10 mg/dL (purged boundary, 301 crossing labels removed); exploratory, not clinical benefit.

Same-fold + workload

Seed-42 held-out comparison: full-fusion HGB PR-AUC 0.382; reduced five-variable EHR + sensors 0.400; persistence 0.308; trend 0.226.

Top-5% episode cutoffs: 199/391 (30m), 197/389 (60m), 122/384 (120m).

30-minute merge and 60-minute cooldown are exploratory; false alerts 0.25–1.14 per approximate participant-day.

Limitations and path to deployment

A research workflow can inform the next study only after local validation, governance, and clinician-defined action.

Current limits

45-participant US cohort; no Indian cohort, external validation, prospective study, fairness assessment, or clinical utility evaluation.

Not a physiological simulator; medication timing/dose, treatment response, sleep, and HRV are not modeled.

Episode merge, cooldown, thresholds, and dashboard scores are retrospective research choices.

Privacy and governance

Assess India Digital Personal Data Protection Act (DPDP) obligations for any local data work.

HIPAA applies only when responsible entities and data fall within its scope; obtain qualified review.

Use consented, governed, de-identified data; do not ship raw records or participant-level derived outputs.

Deployment path

Representative Indian data → local calibration/retraining → prospective multi-site validation.

Define clinician action, escalation, monitoring, audit, and workload targets before any alert workflow.

Only then evaluate secure integration with approved devices and records; this prototype must not connect to care workflows.

Conclusion and submission details

GlycoTwin demonstrates a reproducible multimodal forecasting workflow; the evidence supports further research, not clinical use.

What the prototype shows

Static baseline context plus CGM, heart rate, activity, meals, and time-of-day can be fused at forecast timestamps.

Grouped evaluation, same-fold baselines, chronological personalization, calibration, and episode workload expose trade-offs.

The architecture matches the pipeline: align → fuse 41 dynamic + 12 baseline features (53 total) → HGB → forecast → dashboard/evaluation.

Team

Team name: Sage

Team lead and member: Manda Rohit Vardhan — Project Lead and Developer

College: Amrita Vishwa Vidyapeetham, Hyderabad

Incubator: None

Submission

Demo video: unlisted YouTube link to be added (15–20 minutes)

Repository: github.com/Rohit006-sage/sage-glycotwin-cgmacros

Dataset citation: CGMacros v1.0.0, PhysioNet, DOI 10.13026/3z8q-x658.

Retain the research-only notice and dataset/privacy review before sharing.